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Project Name Concept: Focus IB

Subtitle: Mastering the IB through Deep Work, Not Busy Work.

1. The Problem: The IB "Distraction-Burnout" Loop

IB students face a unique set of pressures: Internal Assessments (IAs), the Extended Essay (EE), CAS, and rigorous exams. Current tools fail them because:

- **Context Switching:** Students jump between ManageBac, Google Docs, YouTube, and Revision Village, losing focus every time.
 - **The Procrastination Paradox:** When work feels overwhelming, students retreat to social media, leading to late-night "grind sessions" that are inefficient.
 - **Generic Scheduling:** Standard calendars don't know the difference between a simple homework assignment and a 4,000-word essay.
-

2. The Solution: A Unified Focus Ecosystem

Focus IB is the first all-in-one platform that combines rigorous focus-enforcement with high-level academic support. It doesn't just tell you *when* to work; it provides the *environment* and *tools* to finish that work faster.

Feature	Description
Rigorous Deep Work	A system-level block that whitelists only the Focus IB website and essential research tools. No social media, no external AI, no distractions.

Smart Flow Scheduling	Connects to school APIs (ManageBac/Toddle) to auto-import deadlines. It schedules tasks based on energy levels (high-brain power for IAs, low-brain power for flashcards).
Integrated Workspace	Write your EE or IAs inside the platform with built-in word counters, IB criteria checklists, and top-tier exemplars visible in the sidebar.
Gamified Accountability	Earn XP and badges for "Deep Work Streaks." Compete on global leaderboards and link with a "Study Buddy" for mutual accountability.

3. Key Product Features

A. The Intelligent Scheduler (AI-Driven)

- **Urgency vs. Weighting:** The platform prioritizes work not just by due date, but by "Summative Weight." An IA draft will always take priority over a minor quiz.
- **Work-Based Rewards:** Instead of a strict 25-minute timer, the "Block" only lifts once a specific milestone is hit (e.g., "Write 300 words" or "Complete 10 Physics Paper 1 questions"). This incentivizes speed and quality.

B. The "Guided" Writing Suite

- **Structural Templates:** Writing a Chemistry IA? The platform splits your screen: left side for writing, right side for the IB Rubric and "Tips for the Evaluation Section."
- **AI Feedback (The Ethical Way):** Our built-in AI doesn't write for you; it **critiques**. It acts like a 24/7 tutor, giving you a "Predicted Grade" based on your current draft and suggesting where you are missing marks in the criteria.
- **Anti-Cheat Environment:** To maintain academic integrity, external AI is blocked, and we provide a built-in "Human-Logic" checker to ensure students are developing their own voice.

C. The Study Vault

- **Active Recall:** A database of past paper questions and flashcards categorized by IB Subject Groups.

- **Tutorials on Demand:** Short, "atomic" video lessons for when a student gets stuck on a specific concept, preventing them from going to YouTube and falling down a rabbit hole.
-

4. The Social & Gamification Engine

"Solitary study is where focus goes to die."

- **Leaderboards:** Students earn "Focus Points" based on efficiency and tasks completed.
 - **Study Buddies:** Users can "lock in" with a friend. If one person breaks their focus block, the other person is notified.
 - **Profile Badges:** Earn "IB Survivor" badges for finishing the EE 48 hours early or "Math King" for 500+ practice questions.
-

5. Why It Wins

1. **Lower Barrier to Entry:** By connecting to school APIs, the student doesn't have to manually "set up" their schedule. It's done for them.
 2. **Psychological Alignment:** By using **Energy-Based Scheduling**, we stop forcing students to do hard math when they are exhausted at 9:00 PM.
 3. **Simplicity:** It replaces the need for a separate Forest app, a separate Notion template, a separate Revision site, and a separate Word processor.
-

6. Specific features

1. Screen Block
 - a. Blocking websites (social medias, video streaming, and AI)
 - b. White listing websites (google scholar, manage bac, ect)
 - c. Time limit or wordcount minimum
 - d. Pomodoro
 - e. Breaks
2. Analytics
 - a. Distraction log
 - b. Focus analytics

- c. Progress visualization
- 3. Scheduler
 - a. tasks/time blocks
 - b. Read students managebac and google classroom emails so it can auto schedule assignments
 - c. Task priority
 - d. Ai scheduler
 - e. Google Calendar/Apple Calendar link
 - f. Due dates
 - g. Subtasks/checklist
 - h. Recurring task
 - i. Task attachments
 - j. Harder tasks when students are most alert and energetic
 - k. Students select what Ib section they are working on
 - l. Smart rescheduling
 - i. **Difficulty estimation (and accounts for time to schedule)**
 - ii. **Procrastination detection**
 - iii. Book working at high energy times
 - iv. For overlapping tasks, due at the same time
 - m. Spaced out studying
 - n. Task classification
- 4. Documents
 - a. Google Docs api
 - b. Google slides api
 - c. Google drive api for saving docs
 - d. Ai feedback
 - e. See a comparison essay
 - f. Premade structures depending on assignments to make it easy
 - g. Criteria checklists
 - h. Break up work based on sections of assignment
 - i. Word count tracker for each part of assignment to make sure there on track
 - j. Tips for each section
 - k. What exactly the criteria is asking
 - l. Examples of what achieving that criterion looks like through pieces of exemplars
 - m. **Auto-save** - Prevent work loss every 30 seconds
 - n. **Version history** - Recover previous document versions
 - o. **Microsoft Office support** -
- 5. Studying
 - a. Past paper questions
 - b. Videos
 - c. Flash cards
 - d. Break down of work (diagrams, essay analysis, math problems, everything)
 - e. Focused practice based on what they are weak on
 - f. Ai feedback on answers

- g. if studying for written things (econ paper 1, 2, or 3, or english essays) break down it by the sections and have them practice each section with tips and feedback)
- 6. AI assistant to ask questions
 - a. Limit API abuse
 - b. Limit it doing school work
 - c. Block copy and pasting
- 7. Sign in
 - a. Create an account
 - b. Link google account
 - c. Link manage bac account
 - d. Sign up
 - e. password/ email
 - f. Setup email verification
- 8. Payment
 - a. Use subscription per year/per month
 - b. 14 Free trial
 - c. Stripe
 - d. Google pay
 - e. Apple pay
 - f. Student discount
 - g. Students refer friends for free months
 - h. School discount(bulk buy)
 - i. Tiered subscriptions to access more features and ai features
- 9. Document storage to access docs
 - a. Use google drive api
- 10. Account management
 - a. Change it to dark mode
 - b. Change pass/email
 - c. Profile icon
 - d. username
- 11. Gamification
 - a. Profile badge(updated for days in a row or amount of hours per week)
 - b. Login streak
 - c. **Leaderboards** (profile badge on leader boards)
 - d. **XP/leveling system(instead of xp do hours of studying)**
- 12. Social & Accountability
 - a. **Study groups/co-working sessions** - Virtual study rooms where friends block distractions together
 - b. **Accountability partners** - Share progress with a friend or parent
 - c. **Leaderboards** - Compare focus time with friends (privacy-conscious)
 - d. **Study session sharing** - "I'm studying Biology for 2 hours" status updates
- 13. Learning
- 14. Content integrity
 - a. AI detection

- b. Compare quotes against sources
- 15. Onboarding
 - a. **Interactive tutorial** - First-time walkthrough of key features
 - b. **Study tips/productivity content** - Blog or resource library
 - c. **AI onboarding assistant** - Helps set up first schedule
 - d. **Quick start templates** - Preset configurations for different study styles
 - e. **Video tutorials** - How to use advanced features
- 16. **Notification system** - Push notifications, email digests of upcoming tasks

App Overview

A comprehensive desktop and mobile productivity application designed for IB and secondary students that combines intelligent screen blocking, AI-powered scheduling, document management, and gamified learning to help students focus and complete their academic work efficiently.

Core Value Proposition

Students can add time-blocked tasks with priorities, and AI automatically schedules their week by integrating with Google Calendar, Apple Calendar, and ManageBac. During focus sessions, the app blocks distracting apps and websites while providing a workspace for writing and studying. Work completion is verified through AI-analyzed word counts (preventing spam), and students earn access back to their blocked content after completing tasks.

Feature Categories

1. Screen Block & Focus System

Core Blocking Features:

- Block specific applications on desktop
- Block websites during focus sessions
- Whitelist approved websites (e.g., Google Scholar, research databases)
- Whitelist approved applications needed for schoolwork

- Time-based limits OR word count-based limits for task completion
- AI word count reviewer (detects spam/gibberish to prevent cheating the system)

Focus Management:

- Pomodoro timer integration for structured work/break cycles
- Automated break reminders between focus sessions
- Focus mode intensity levels (adaptive recommendations: if students don't follow soft limits, suggest harder blocking)
- Emergency override system (limited to 1 per session, requires written explanation/reason)

Analytics & Insights:

- Distraction log (tracks attempts to access blocked sites/apps)
 - Focus analytics (total time focused, productivity scores, trends)
 - Progress visualization (completion percentages, streaks, weekly summaries)
-

2. AI-Powered Scheduler

Task Management:

- Create tasks/time blocks with titles and descriptions
- Set task priorities (high/medium/low)
- Add due dates for assignments
- Break tasks into subtasks/checklists
- Set recurring tasks (daily/weekly homework)
- Attach files, rubrics, or reference materials to tasks
- Students select which IB subject section they're working on

Calendar Integration:

- Google Calendar API sync
- Apple Calendar sync
- ManageBac API integration to import class schedule and assignments
- View combined schedule from all sources

Smart AI Scheduling:

- Automatically schedule tasks based on priority, due dates, and available time
- Schedule harder/complex tasks during times when students report high energy/alertness
- Implement spaced repetition for studying (spread out review sessions)
- Task classification by subject and type
- Smart rescheduling capabilities:

- Difficulty estimation (AI predicts time needed based on task type and past performance)
 - Procrastination detection (notifies avoidance patterns and intervenes)
 - Prioritize booking high-energy time slots for demanding work
 - Resolve conflicts when multiple tasks are due simultaneously
 - Auto-adjust schedule when tasks run over or complete early
-

3. Document Management & Creation

Document Integration:

- Google Docs API (create/edit documents in-app)
- Google Slides API (presentation support)
- Google Drive API (save and retrieve documents)
- Microsoft Office support (Word, PowerPoint, Excel)
- Auto-save every 30 seconds to prevent work loss
- Version history (recover previous document versions)

AI Writing Support:

- AI feedback on writing (grammar, structure, clarity)
 - Compare essay versions side-by-side
 - AI assistant for answering questions (with limitations)
 - API abuse limits to prevent overuse
 - Restrictions to prevent AI from doing schoolwork for students
 - Teacher-configurable AI usage limits
 - Block copy-pasting AI-generated content directly
-

4. Content Integrity & Academic Honesty

Verification Systems:

- Revision history tracking (shows authentic work process over time)
 - AI detection via Turnitin API integration
 - Source checker (validates in-text citations by comparing quotes to provided sources, catches fake citations)
-

5. Learning Tools

Study Support:

- Reading comprehension quizzes (verify students read assigned materials)
 - Study material generator (create flashcards or practice questions from notes)
 - IB section-specific practice questions
-

6. Authentication & Account Management

Sign In/Sign Up:

- Create account with email/password
- Email verification during setup
- Link Google account for easy sign-in
- Link ManageBac account for schedule import

Account Settings:

- Change password and email
 - Profile icon and username customization
 - Dark mode toggle
 - Export personal data (documents and analytics)
 - Multi-device sync across desktop and mobile
-

7. Payment & Subscription Model

Pricing Structure:

- Monthly and yearly subscription options
- 14-day free trial for new users
- Tiered subscription plans (more features and AI usage at higher tiers)
- Student discount pricing
- School/institutional bulk purchase discounts
- Referral program (students refer friends for free months)

Payment Processing:

- Stripe integration
 - Google Pay support
 - Apple Pay support
-

8. Gamification & Motivation

Reward Systems:

- Profile badges (updated for consecutive days or hours per week)
 - Login streak tracking
 - Hours-based leveling system (levels increase with study hours, not XP)
 - Leaderboards comparing focus time with friends (privacy-conscious, opt-in)
 - Profile badges displayed on leaderboards
-

9. Social & Accountability Features

Collaborative Study:

- Virtual study rooms (friends can block distractions together in co-working sessions)
- Accountability partners (share progress with friends or parents)
- Study session sharing (broadcast "I'm studying Biology for 2 hours" status)

Oversight Options:

- Parent/teacher dashboard (view student progress without controlling account)
 - Optional leaderboards for friendly competition
-

10. Wellness & Health Integration

Health Reminders:

- Sleep schedule integration (warn against late-night scheduling)
 - Hydration reminders during long study sessions
 - Posture check reminders
-

11. Miscellaneous Features

Entertainment & Breaks:

- YouTube Music API embed (students can listen to music during focus)
- Pong/Tetris games available during Pomodoro breaks
- Typing challenges during breaks

Unique Features:

- Financial self-accountability (optional: miss a deadline, donate \$10 to charity)
 - Multi-device sync (desktop and mobile companion app)
 - Accessibility features (screen reader support, dyslexia-friendly fonts, colorblind modes)
 - API abuse limit strategy across all AI features
-

12. Mobile Companion App

Core Mobile Features:

- Calendar view and sync
 - Screen blocking for phone during study sessions
 - Basic task viewing and check-off
 - Focus session start/stop
 - Notifications
-

13. Notifications & Reminders

Alert System:

- Push notifications for upcoming tasks
 - Email digests of daily/weekly schedule
 - Break reminders during focus sessions
 - Task deadline warnings
-

14. Data Privacy & Compliance

Security & Privacy:

- End-to-end encryption for student documents
 - FERPA compliance (required for US schools)
 - Data export functionality (students can download all their data)
 - Transparent data collection and usage policies
-

15. Onboarding & Support

User Onboarding:

- Interactive tutorial walkthrough for first-time users

- AI onboarding assistant (helps set up first schedule and preferences)
 - Quick start templates (preset configurations for different study styles)
 - Video tutorials for advanced features
 - Study tips and productivity content (blog/resource library)
-

Target Users

- Primary: IB (International Baccalaureate) students
- Secondary: High school and college students
- Institutional: Schools purchasing bulk licenses

Technical Architecture

- **Desktop:** macOS and Windows applications
- **Mobile:** iOS and Android companion apps
- **Integrations:** Google Workspace, Apple Calendar, ManageBac, Turnitin, Microsoft Office
- **APIs:** Stripe payments, OAuth for authentication

Key Differentiators

1. IB-specific features and ManageBac integration
2. AI scheduler that accounts for energy levels and learning science
3. Content integrity tools (source checker, revision tracking)
4. Balanced AI assistance (helpful but prevents academic dishonesty)
5. Holistic approach combining blocking, scheduling, writing, and wellness

-App layout

- Find similar study, planner/scheduler, screen time blocking apps/websites that you enjoy using or have a good design or any website you like at all and link them here also highlight there main strengths

-also paste the apps/websites you don't like using and what about them sucks

<https://superhuman.com/>

1. Freedom

- a. Strengths
 - i. Blocks both websites and apps across multiple devices (computer, phones, tablets)
 - ii. Locked mode and uninstall protection mode - not letting the users to log out from the app or uninstall the app so that they are forced to study
 - iii. Sound system - focus music instilled in the app
- b. Weakness
 - i. It requires a subscription to unlock every feature (\$8 a month), not completely free.
 - ii. Blog section - not necessary annoying

2. Forest

- a. Strengths
 - i. The gamification is great - it basically has an inbuilt game where you have to focus, and your tree grows along with the time you focus on your tree and making the forest richer. gives a psychological boost to users, allowing them to maintain their focus.
 - ii. Interesting UI, easy to start, and it's almost identical to a gaming app
- b. Weaknesses
 - i. no focus tools like music or pomodoro functions.
 - ii. It is not dedicated to education, so while blocking apps the users can do whatever they want, might lose their focus
 - iii. Mostly limited to mobile/phone usage - not blocking PC/Web
 - iv. No scheduling feature

3. MyStudyLife

- a. Strengths
 - i. Dedicated to academic life: supports rotating schedules, tracking exams, and assignments
 - ii. Has a clear indication to motivate the users: 4 categories
 - 1. Pending tasks
 - 2. Overdue tasks
 - 3. Tasks completed
 - 4. Your streak

- iii. Cross-platform sync - from mobile to web, from web to mobile
- b. Weaknesses
 - i. It doesn't provide any app blocking
 - ii. Doesn't have advanced AI scheduling or prediction features

-Font choice

-Colour choice

<https://mobbin.com/discover/apps/ios/latest>

(avoid using the colours of websites that are similar to ours)

https://mobbin.com/apps/how-we-feel-ios-88680053-11ec-4c6c-8763-1e2aa70388e7?utm_source=copy_link&utm_medium=link&utm_campaign=app_sharing

- a. Base color black -> It reduces eyestrain and conveys a calm and focus overall atmosphere.
- b. Accent yellow or any bright colors -> Creates urgency and draws attention to the main buttons. And could be used for positive reinforcements ("completed", "365-day streak!")
- c. Overall - More energetic and motivational look

https://mobbin.com/apps/open-ios-c244e38f-ae98-4b62-aa01-12efbcc3715f?utm_source=copy_link&utm_medium=link&utm_campaign=app_sharing

- a. Base color black -> More calm and professional looking. Minimize distractions.
- b. Accent color -> brings attention
- c. Lower saturation -> feel more natural
- d. Overall - more natural and grounded atmosphere

https://mobbin.com/apps/luma-ios-bead4230-994f-47c2-9311-5049bf7bcace?utm_source=copy_link&utm_medium=link&utm_campaign=app_sharing

- a. Base color white -> simple, not too noisy, emphasizes other components of the app
- b. Accent color = pink and purple -> more interactive, more like modern SNS communication app vibes.
- c. Overall - more interactive, additional emphasis on interactive & accountability feature of the app

<https://tryhero.app/>

<https://tryhero.app/>

-Tech stack

Frontend Framework

Electron - Best choice for your use case because:

UI Framework

React + TypeScript

Styling: Tailwind CSS + shadcn/ui

Backend

Node.js + NestJS

Database

PostgreSQL (primary)

Redis (caching layer)

AI & ML

Kimi K2 or what ever Cheap Ai is available

- Writing feedback and AI assistant features
-

LangChain (for managing Ai

Cloud Infrastructure

AWS or Google Cloud Platform

- AWS recommended for mature services
- S3 for document storage
- Lambda for serverless functions (document processing, AI tasks)
- RDS for managed PostgreSQL

Deployment: Docker + Kubernetes (or AWS ECS for simpler management) (for testing)

Key Integrations

Authentication: Appwrite

Payment: Stripe

Calendar:

- Google Calendar API (o
- Apple CalDAV protocol
- ManageBac API (custom scraping likely needed)

Document APIs:

- Google Workspace APIs (Docs, Slides, Drive)
- Microsoft Graph API (Office 365)

Content Integrity:

- Turnitin API (requires institutional partnership)
- Custom plagiarism detection as fallback

Native System Access

node-window-manager (for blocking)

Custom Native Modules:

- May need Rust or C++ modules for deep system integration
- Platform-specific APIs for reliable blocking (harder to bypass)

Mobile (Future)

React Native

Alternative:

Development Tools

- Version Control: Git + GitHub
- CI/CD: GitHub Actions
- Monitoring: Sentry (error tracking), LogRocket (session replay)

- Analytics: Mixpanel or Amplitude
- Testing: Jest, React Testing Library, Playwright (E2E)

Blocking

- Windows Filtering Platform (WFP) for network filtering
- Windows Driver Kit if you need kernel-level blocking (most secure)

Mac:

- Swift for native apps with System Extension
- Network Extension framework for web filtering
- Endpoint Security framework for app blocking

Preventing Deletion/Bypass

Windows:

- Install as a Windows Service (runs in background, harder to stop)
- Use Group Policy to prevent uninstallation (requires admin)
- File system protection: Mark files as system/hidden, use file locking
- Monitor your own process and restart if terminated
- Password-protect uninstallation within your app

Mac:

- System Extensions run at system level and are harder to remove
- Require admin password for uninstallation
- Use Launch Daemon (runs at boot, system-level)
- Code signing and notarization (makes it look legitimate to OS)
- Password-protect settings changes
-

-Who are our competitors

Hero is our main competitor their website is good looking but they don't offer screen blocking and document editing so for that we are unique and absolutely no one has created any app for the IB

-How are we gonna market

We are going to create organic content on instagram, youtube shorts, and tiktok as well as promoted content on snapchat spotlight and facebook

-what is our niche

-what is our brand identity

-who is our demographic

-what is our user flow

-who can we take inspiration from

-what are our biggest risks

-how do we prevent those risks

-how do we use AI

-how do we manage version control

-what platforms are we gonna use

-what MCP's, API's, and AI's will we use

BMAD

GLM 4.6

GBT 5 codex

Droid

-What don't we know

-how do we build this

Analyze these videos

<https://www.youtube.com/watch?v=ACeGhRCg4Go>

<https://www.youtube.com/watch?v=VZkm1jSs8Lg>

<https://www.youtube.com/watch?v=wiQMsQwWJQE>

https://www.youtube.com/watch?v=fg0_0M8kZ8g

<https://www.youtube.com/watch?v=JVwkyYtZNzQ>

1. The Visual Palette: "Intellectual High-Contrast"

To capture the "locked-in" energy, we are moving away from the soft, friendly pastels of typical productivity apps and leaning into high-stakes, high-contrast colors.

- Primary: "Oxford Yellow" (#FFD700 / #FFC107)
 - *Psychology*: This isn't a "happy" yellow; it's the yellow of a highlighter on a textbook, a caution sign, and mental clarity. It represents high energy and alertness.
- Secondary: "Inkwell Black" (#0A0A0A)
 - *Psychology*: Deep, absolute black. It represents the "void" of deep work where distractions disappear. It provides the "solid" feel of a premium fountain pen or an old-school chalkboard.
- Accent: "Paper White" (#F5F5F5)
 - *Use*: For UI backgrounds and text to maintain readability while keeping the "analog" academic feel.

Typography Stack

- Headlines (Primary): **Crimson Pro (Semi-Bold)**
 - *Energy*: High-contrast, editorial, prestigious.
- Interface/Body (Secondary): **Spectral (Regular/Medium)**
 - *Energy*: Elegant, non-technical, academic clarity.
- Data & System Stats (Tertiary): **Courier Prime**
 - *Energy*: The "Grind" font. Used for timers, word counts, and XP totals.

3. Target Audience: "The Ambitious Overwhelmed"

- **The Persona**: High-achieving 16–19 year olds who are academically gifted but drowning in the "IB Logistics."
- **Their Goal**: To regain their weekends. They don't want to work *more*; they want to work *better* so they can actually sleep.
- **Brand Relationship**: Focus IB isn't a "cheerleader"; it's a **strict but brilliant mentor**. It tells them exactly what to do and shuts the door on distractions until the job is done.

Problem examples

- Extended essay
- tok
- cas
- cas project

Issues

- After the first two parts it takes too long
- IB specific examples for pop ups and for questions
- We want them to read the examples
- Typing part is way too long
- too much space on navigation bars
- make the cas, tok, exam part
- maybe cut out the typing part after the scheduling

And say something like other problems IB students run into

- make typing a bit faster
- keep font consistent